

Post-Bin Determinant Energy on the Actual Support: Exact Binning SVD, Kernel Angle, and a Natural-Scale Obstruction

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Abstract

Let $\mathcal{T}_X$ be the canonical parent archive of the literal fixed- h_0 matched coefficient, let $\nu : \mathcal{T}_X \rightarrow \mathcal{N}_X$ be its determinant map, and write

$$A_X(n) = \sum_{\nu(t)=n} u_t, \quad D_X = \sum_n |A_X(n)|^2.$$

We analyze this *post-bin* energy on its actual finite support, without replacing the coefficient by an ambient or random model. The normalized determinant-binning operator is a coisometry. Its adjoint projection is exact averaging on every determinant fiber, and its kernel is the direct sum of the zero-sum subspaces of those fibers. If Π_X denotes this projection and m_n the actual fiber multiplicity, then

$$D_X = \sum_n m_n \|(\Pi_X u)_{F_n}\|_2^2.$$

Thus determinant-energy survival is exactly a weighted angle problem between the literal coefficient and the binning kernel.

This gives a sharp structural dichotomy. If all determinant fibers are singletons, then D_X equals the pre-bin diagonal energy, whose imported upper envelope is one full power below the desired natural scale. If collisions occur, the support map has a nontrivial kernel and support or occupancy alone supplies no lower angle. Therefore any lower bound with loss $\lambda_D < 1$ must use positive equal-determinant off-diagonal coherence of the actual coefficient. We give exact necessary and sufficient certificate statistics, but prove no growing arithmetic lower bound. The result is an L0/L1 obstruction and interface theorem, not an L2 parity advance.

Keywords: determinant binning; kernel angle; equal-determinant coherence; fixed shift.

1 Literal packet and the question

Fix one prescribed $h_0 \in \mathbb{Z} \setminus \{0\}$. In the canonical parent archive of TPC-95, a native key $t \in \mathcal{T}_X$ carries its literal coefficient u_t and determinant $\nu(t) \neq 0$ [4]. Computational children have already been inverse-aggregated, so duplicates are not counted as new physical keys. Put

$$F_n = \{t \in \mathcal{T}_X : \nu(t) = n\}, \quad m_n = |F_n|, \quad A_X(n) = \sum_{t \in F_n} u_t.$$

Only nonempty fibers occur in $\mathcal{N}_X = \nu(\mathcal{T}_X)$.

The two energies are

$$E_X = \sum_{t \in \mathcal{T}_X} |u_t|^2, \quad D_X = \sum_{n \in \mathcal{N}_X} |A_X(n)|^2.$$

The imported scale ledger is

$$Q_X = X^{267/400+o(1)}, \quad J_X = X^{133/400+o(1)}$$

and

$$E_X \ll X^{-1+o(1)} Q_X^3 J_X^2.$$

The open natural-scale target has the form

$$D_X \geq X^{-\lambda_D+o(1)} Q_X^3 J_X^2$$

with, at minimum, a fixed $\lambda_D < 1$ [1–3].

2 Exact SVD of determinant binning

Let $H_X = \mathbb{C}^{\mathcal{T}_X}$, $K_X = \mathbb{C}^{\mathcal{N}_X}$, both with counting measure. Define the raw and normalized binning maps

$$(B_X v)_n = \sum_{t \in F_n} v_t, \quad (U_X v)_n = \frac{1}{\sqrt{m_n}} \sum_{t \in F_n} v_t.$$

Theorem 2.1 (Actual-support binning SVD). *The normalized map satisfies*

$$U_X U_X^* = I_{K_X}.$$

Consequently $\Pi_X = U_X^* U_X$ is the orthogonal projection onto the subspace of vectors constant on each determinant fiber, and

$$(\Pi_X v)_t = \frac{1}{m_n} \sum_{s \in F_n} v_s \quad (t \in F_n).$$

Moreover,

$$\text{Ker } B_X = \text{Ker } U_X = \bigoplus_{n \in \mathcal{N}_X} \left\{ v \in \mathbb{C}^{F_n} : \sum_{t \in F_n} v_t = 0 \right\},$$

so

$$\dim \text{Ker } B_X = |\mathcal{T}_X| - |\mathcal{N}_X|.$$

Proof. For $w \in K_X$, the adjoint has

$$(U_X^* w)_t = \frac{w_n}{\sqrt{m_n}} \quad (t \in F_n).$$

Summing this over F_n and dividing by $\sqrt{m_n}$ returns w_n , proving $U_X U_X^* = I$. The projection formula follows. The kernel statements are the defining zero-sum conditions, one independent equation per nonempty fiber. $\square$

Corollary 2.2 (Exact post-bin energy identity). *For the literal coefficient $u = (u_t)$,*

$$D_X = \sum_{n \in \mathcal{N}_X} m_n \|\Pi_X u|_{F_n}\|_2^2.$$

If the actual carrier is empty, then $D_X = 0$ and the following extrema are omitted. Otherwise, writing

$$m_{\min} = \min_n m_n, \quad m_{\max} = \max_n m_n,$$

one has

$$m_{\min} \|\Pi_X u\|_2^2 \leq D_X \leq m_{\max} \|\Pi_X u\|_2^2.$$

Proof. On F_n , the projected vector has constant value $A_X(n)/m_n$, hence its squared norm is $|A_X(n)|^2/m_n$. Multiply by m_n and sum. $\square$

3 The exact kernel-angle certificate

Definition 3.1 (Actual alignment statistics). For $u \neq 0$, define

$$\alpha_X = \frac{\|\Pi_X u\|_2^2}{\|u\|_2^2} \in [0, 1].$$

If $\Pi_X u \neq 0$, define its effective multiplicity

$$\overline{m}_X = \frac{D_X}{\|\Pi_X u\|_2^2}.$$

If $\Pi_X u = 0$, set $\overline{m}_X = 0$.

Proposition 3.2 (Necessary and sufficient energy statistic). *For every nonzero literal coefficient,*

$$\boxed{D_X = E_X \alpha_X \overline{m}_X},$$

where either $\alpha_X = \overline{m}_X = 0$, or

$$m_{\min} \leq \overline{m}_X \leq m_{\max}.$$

Therefore a proposed lower bound for D_X is equivalent to the same lower bound for the product $E_X \alpha_X \overline{m}_X$.

Proof. The identity is the definition of $\overline{m}_X$, with $\|\Pi_X u\|_2^2 = E_X \alpha_X$. The range follows from [corollary 2.2](#). $\square$

Remark 3.3 (Not a circular proof device). [Proposition 3.2](#) is a diagnostic factorization. It is useful only if E_X , α_X , and $\overline{m}_X$ are estimated by independent literal arguments. Renaming D_X/E_X as an angle is not an arithmetic lower bound.

Corollary 3.4 (Forced amplification below loss one). *Suppose the imported upper envelope for E_X holds and, as an additional future theorem,*

$$D_X \geq X^{-\lambda_D + o(1)} Q_X^3 J_X^2 \quad \text{for fixed } \lambda_D < 1.$$

Then

$$\boxed{\alpha_X \overline{m}_X = \frac{D_X}{E_X} \geq X^{1-\lambda_D - o(1)}}.$$

Proof. Divide the assumed lower bound by the imported upper bound for E_X . $\square$

4 Sharp structural obstruction

Theorem 4.1 (Singleton-or-kernel dichotomy). *Exactly one of the following holds.*

(i) *Every actual determinant fiber is a singleton. Then*

$$D_X = E_X \ll X^{-1+o(1)} Q_X^3 J_X^2,$$

so the diagonal route alone cannot prove a lower bound with fixed $\lambda_D < 1$.

(ii) *At least one actual determinant fiber has multiplicity at least two. Then B_X has a nontrivial zero-sum kernel on the actual carrier, and the support partition alone gives no positive lower bound for α_X .*

Proof. In the singleton case B_X is a coordinate relabeling, so $D_X = E_X$, and the imported diagonal envelope applies. In the collision case, choose two coordinates in one fiber with values 1, -1 and all other coordinates zero. This nonzero vector lies in $\text{Ker } B_X$, so its alignment is zero. Hence no positive lower angle follows from the partition alone. $\square$

Proposition 4.2 (Full-support sharpness). *Assume at least one collided fiber. Among coefficient vectors nonzero on every carrier coordinate, the ratio*

$$\frac{D(v)}{\|v\|_2^2}$$

can approach zero whenever every fiber is collided; if singleton fibers also occur, the contribution of those fibers can be made arbitrarily small relative to a zero-sum collided component. Thus merely declaring every carrier key active does not create a uniform lower angle.

Proof. On each collided fiber use nonzero roots of unity, whose sum is zero. If singleton fibers occur, assign them a common amplitude ε and let $\varepsilon \rightarrow 0$, while retaining a fixed zero-sum component on one collided fiber. $\square$

5 Off-diagonal coherence is unavoidable

Define the ordered equal-determinant coherence

$$C_{\text{eq},X} = \sum_n \sum_{\substack{t,s \in F_n \\ t \neq s}} u_t \overline{u_s}.$$

Then the exact identity is

$$D_X = E_X + C_{\text{eq},X}.$$

Theorem 5.1 (Necessary coherent main term). *Under the hypothetical lower bound in [corollary 3.4](#), with fixed $\lambda_D < 1$,*

$$C_{\text{eq},X} = (1 - o(1))D_X.$$

In particular, any successful below-loss-one energy theorem must produce asymptotically positive equal-determinant off-diagonal coherence on the actual coefficient.

Proof. The imported diagonal envelope and the hypothetical lower bound give

$$\frac{E_X}{D_X} \ll X^{\lambda_D - 1 + o(1)} = o(1).$$

Use $C_{\text{eq},X} = D_X - E_X$. $\square$

TPC-95 proves that complete shared-target collisions are already lower order under any such future energy theorem [4]. Therefore a successful coherent main term would have to lie in the remaining determinant-preserving structure, not in duplicated native targets.

6 Audited conclusion

Statement	Status
Actual-support binning SVD and kernel	Exact finite theorem (L0); literal identification is L1.
Natural-scale lower bound for D_X	Not proved.
Support-only determinant-energy route	Stopped by the singleton-or-kernel dichotomy.
Required new input	Arithmetic control of the literal kernel angle or positive equal-determinant coherence.
Distinguished zero mode	Not estimated here; TPC-111 treats it independently.
Parity or prime-pair consequence	None; no L2 claim.

The paper establishes the strongest support-only statement available: the complete SVD, the exact diagnostic product, and a sharp obstruction. It does not show that the actual coefficient is close to the binning kernel, nor that it stays away from it.

Independent ledgers. Even a future determinant exponent λ_D must be paired with an independently proved zero-mode exponent η_Z satisfying $\lambda_D \leq 2\eta_Z$. That compatibility is separate from the physical synthesis condition $\Lambda_{\text{phys}} < 1/400$. No loss in this paper is credited to both ledgers.

Explicit exclusions. No result is specialized to $h_0 = 2$. This paper proves no Möbius or Liouville cancellation, no determinant-energy lower bound at a growing scale, no zero-mode estimate, no breach of the sieve parity barrier, no Hardy–Littlewood asymptotic, and no twin-prime theorem.

References

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